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LARGE-SCALE URBAN SCENE RECONSTRUCTION USING 3D GAUSSIAN SPLATTING AND SATELLITE IMAGERY

Hanqing Xu
East China Normal University
500 Dongchuan Rd, ShangHai, China
51273901140@stu.ecnu.edu.cn

Lingfei Ma*
East China Normal University
500 Dongchuan Rd, ShangHai, China
mlfcufe@163.com

Haiyan Guan
Nanjing University of Information Science & Technology
219 Ningliu Rd, Nanjing, China
guanhy.nj@nuist.edu.cn

Jonathan Li
University of Waterloo
200 University Ave W, Waterloo, Canada
junli@uwaterloo.ca

***Abstract*—With the rapid development of deep learning techniques, the construction of digital twin cities has become an important direction for developing digital earth. Thus, in this paper, we perform scene reconstruction using the 3D Gaussian Splatting (3DGS) method and satellite imagery in complex and large-scale urban downtown areas. Experimental results show that 3DGS significantly outperformed Neural Radiance Fields (NeRF) in both reconstruction accuracy and efficiency, improving approximately 10–13 dB in PSNR, 0.3–0.4 in SSIM, and a reduction of 0.1 in LPIPS. Moreover, 3DGS exhibited high robustness under multi-period lighting conditions, with most quantitative metric fluctuations remaining below 2%. This paper highlights the potential of 3DGS in panoramic urban modeling and multi-scene analysis for digital twins, providing efficient and reliable technical support for future urban planning and management.**

***Index Terms*—3D Gaussian splatting, neural radiance fields, large-scale 3D modeling, digital twins.**

I. Introduction

In recent years, the quick advancement of deep learning and satellite techniques has promoted the applications of smart cities and digital twins [1][2]. Creating high-definition digital twins could enable virtual-real interaction and real-time mapping to develop digital and intelligent transformation and upgrading of urban construction. Current 3D city modeling technologies are primarily divided into two categories: (1) interactive modeling based on data sources, such as Computer-Aided Design (CAD) and Building Information Modeling (BIM) [3]; and (2) semi-automatic modeling using optical imagery or 3D point clouds obtained via laser scanning systems, the latter being the mainstream for large-scale urban scene construction[4].

Accordingly, 3D reconstruction methods based on Neural Radiance Fields (NeRF) [5] and 3D Gaussian Splatting (3DGS) [6] have gained significant attention. NeRF uses multilayer perceptrons (MLPs) [7] to learn spatial color and geometric density, generating high-quality views through differentiable volume rendering [8]. However, NeRF excels in lighting conditions and shadow details, its implicit representation leads to low computational efficiency and longer training times. In contrast, 3DGS adopts an explicit geometric representation, initializing sparse point clouds as 3D Gaussian ellipsoids [9]. Complex scenes are rendered using splatting techniques and optimized through differentiable Gaussian distributions, enabling 3DGS to directly output geometric structures with higher efficiency and flexibility, and making it particularly suited for diverse applications such as urban modeling, autonomous driving, and augmented reality (AR) [10]. However, in practical applications, the lighting conditions of scenes may change significantly over time (e.g., early morning, noon, and dusk), and such lighting differences often affect the effectiveness and accuracy of traditional 3D reconstruction methods [11].

To solve this problem, we collected satellite imagery from different times to explore the impact of lighting changes on the reconstruction performance of 3DGS and verify its robustness in complex lighting scenarios. More specifically, this study utilized multi-view satellite images obtained from Google Earth Studio to train a 3DGS model, achieving effective and accurate scene reconstruction in complex urban downtown areas. The main contributions of this paper are threefold as follows: (1) We investigate 3DGS performance under multi-temporal lighting in complex urban scenes, confirming its robustness to illumination changes; (2) We compare 3DGS and NeRF in three dense urban landmarks, demonstrating higher quality and over 10× faster training with 3DGS; and (3) We propose a scalable 3DGS-based pipeline using satellite imagery from Google Earth Studio for practical digital twin modeling.

*Corresponding author: mlfcufe@163.com (Lingfei Ma)

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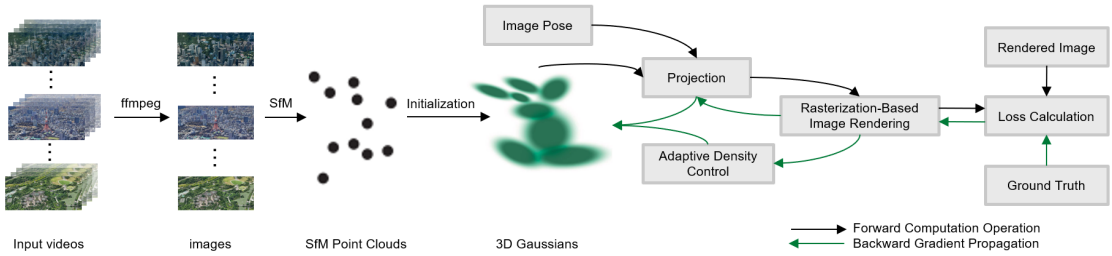

Fig. 1. 3D Gaussian Splatting workflow.

II. Method

A. SfM Point Cloud Initialization

The implementation of 3D Gaussian Splatting (3DGS) relies on pre-processing with COLMAP-based [12] Structure-from-Motion (SfM) technology [13]. SfM is a technique that estimates 3D structures from multiperspective images, capable of taking a set of unordered images with unknown camera poses as inputs to generate corresponding camera poses and a sparse point cloud. The point cloud initialized through SfM not only maintains the ideal characteristics of continuous radiance fields, providing a solid foundation for subsequent scene optimization, but also effectively reduces redundant calculations in empty areas, thereby enhancing optimization efficiency. In contrast, although 3DGS supports random point cloud initialization, it significantly slows down the optimization convergence rate and substantially extends training time. Therefore, COLMAP-based SfM preprocessing achieves a better balance between efficiency and effectiveness.

B. 3DGS

Fig. 1 illustrates the overall workflow to create high-definition city models using 3DGS, which mainly consists of three modules: 3D Gaussian Ellipsoid initialization, scene redering and optimization, and densification and cropping.

1) Initializing a Set of 3D Gaussian Ellipsoids: Sparse point cloud points are initialized as 3D Gaussian ellipsoids. Each ellipsoid is defined by a Gaussian distribution as follows:

$$G(x) = e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} \tag{1}$$

where μ represents the 3D coordinates of the point, and Σ is the covariance matrix, which defines the shape and orientation of the Gaussian distribution.Here, $G(x)$ represents the probability density of a 3D point x in the Gaussian distribution. The covariance matrix is expressed as:

$$\Sigma = RSS^T R^T \tag{2}$$

where R is the rotation matrix (quaternion) that controls the orientation, and S is the diagonal matrix that defines the length, width, and height of the ellipsoid. Color and lighting are calculated using spherical harmonics by Eq. (3):

$$C = \sum_{i=1}^{N} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \tag{3}$$

where c_i and α_i denote the color and opacity, respectively, and α_j represents the opacity of the j-th point.

2) Scene Rendering and Optimization: Through rasterization, 3D ellipsoids are rendered. The covariance matrix is projected into 2D using the following equaiton:

$$\Sigma' = JW\Sigma W^T J^T \tag{4}$$

where W and J are transformation matrices in the rendering pipeline. Then, the projected Gaussian distribution is used to generate pixel values, while pixel colors are calculated through depth sorting and alpha blending as follows:

$$\alpha_i(x) = a_i G_{2D}(x) \tag{5}$$

where $G_{2D}(x)$ is the 2D Gaussian distribution, and a_i is the opacity of the point. The generated view is compared with the ground truth image, and the image quality is evaluated using the following loss function:

$$L = (1 - \lambda)L_1 + \lambda_{\text{D-SSIM}} \tag{6}$$

where L_1 measures the absolute pixel error, D-SSIM evaluates the structural similarity difference, and λ is a weight parameter with its default value of 0.2 in this paper.

3) Densification and Cropping: 3DGS enhances scene details through dynamic densification and pruning mechanisms. In complex areas with insufficient geometric detail, new Gaussian ellipsoids are generated, which low-variance ellipsoids are duplicated, while high-variance ellipsoids are split into smaller sub-ellipsoids. At the same time, ellipsoids with opacity below the threshold $\epsilon_a = 5 \times 10^{-3}$ are pruned to remove redundancies. This mechanism optimizes the geometric distribution while effectively controlling the total number of ellipsoids, significantly improving computational efficiency.

C. Evaluation metrics

To assess the quality of synthesized images, we employ Peak Signal-to-Noise Ratio (PSNR) [14], Structural Similarity Index (SSIM) [15], and Learned Perceptual Image Patch Similarity (LPIPS) [16] as comprehensive evaluation metrics to compare generated views with ground truths. PSNR and SSIM assess pixel-level distortion and structural similarity, respectively, while LPIPS captures perceptual similarity using deep neural features. Higher PSNR/SSIM and lower LPIPS indicate better reconstruction quality.

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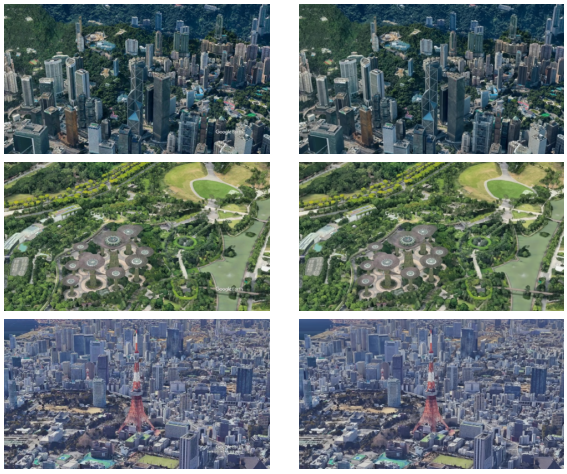

Ground Truth Reconstruction Results

Fig. 2. Urban scene reconstruction results using 3DGS. Top: Bank of China Tower, Hong Kong; Middle: Gardens by the Bay, Singapore; Bottom: Tokyo Tower, Japan.

III. Dataset and Implementation Details

The study areas include three metropolitan cities: Hong Kong, Singapore, and Tokyo, centered on the Bank of China Tower (22.293°N, 114.161°E), Gardens by the Bay (1.287°N, 103.864°E), and Tokyo Tower (35.67°N, 139.745°E), respectively. Each study area covers an approximate area of 80 hectares.

TABLE I
RECONSTRUCTION RESULT COMPARISON BETWEEN NERF AND 3DGS (PSNR IN dB, SSIM AND LPIPS DIMENSIONLESS)

Dataset	Metrics	NERF	3DGS
Bank of China Tower	PSNR	22.566	34.1
	SSIM	0.546	0.980
	LPIPS	0.1764	0.0386
Gardens by the Bay	PSNR	24.296	35.6
	SSIM	0.730	0.985
	LPIPS	0.1058	0.0224
Tokyo Tower	PSNR	21.094	32.6
	SSIM	0.639	0.968
	LPIPS	0.1066	0.0436

At the urban scale, we used Google Earth Studio to define concentric camera paths with varying radii and altitudes centered on the three landmarks, generating multi-perspective aerial and satellite images. Each 15-second rendered video produced 451 frames at a resolution of 960×540 pixels. All experiments in this study were conducted on a desktop computer equipped with an Intel(R) Core(TM) i5-14600KF CPU, NVIDIA GeForce RTX 4070 SUPER (12G) GPU, 32GB RAM, Python version 3.9.10, and CUDA toolkit version 11.8.

IV. Experiments and Results

The reconstruction results demonstrated that 3DGS efficiently generated 3D urban scene models with rich details and clear geometric structures, achieving outstanding performance. As shown in Table 1, in the Bank of China Tower dataset, 3DGS achieves a PSNR of 34.1 dB, and the PSNR across all datasets exceeds 30 dB, indicating high image quality and low noise levels. Furthermore, SSIM is close to 1, and LPIPS is below 0.05, further validating that the rendered images by 3DGS are highly consistent with real-world ground truth images in terms of structural similarity and perceptual quality. As shown in Figure 2, the rendered images are visually indistinguishable from the real-world ground truth images, with only slight differences in the watermark areas.

To further evaluate the performance of the 3DGS method, we designed a series of comparative experiments, comparing it with the well-known NeRF method in recent years. Both methods were trained on the same dataset.

As shown in Table 1, 3DGS significantly outperforms NeRF in both reconstruction accuracy and computational efficiency. Across all datasets, 3DGS achieves a PSNR improvement of 10-13 dB, an increase in SSIM of 0.3-0.4, and a reduction in LPIPS of 0.07-0.13, demonstrating superior image quality and geometric fidelity. Under the same 30,000 iterations, 3DGS finished training in approximately 1.5 hours, while NeRF took over 16 hours on the same hardware. This more than 10× improvement in efficiency, combined with better reconstruction quality, highlights the practicality of 3DGS for large-scale digital twin modeling. Furthermore, 3DGS does not require additional spherical parameterization (spherify) to correctly process 360° surround-shot datasets, whereas NeRF should perform this operation to properly render similar scenes, further validating the broad applicability and operational simplicity of 3DGS across different scenarios.

However, a major challenge in urban modeling is the variation in lighting conditions, which often affects the stability and accuracy of 3D reconstruction methods. To address this issue, we utilized different satellite image data from three time periods, i.e., early morning, noon, and dusk, to evaluate the performance of the 3DGS method under different lighting conditions. Fig. 3 illstrates the urban scene reconstruction results under various lighting conditions in the Singapore dataset. Experimental results showed that, whether under low light conditions (early morning, dusk) or strong light conditions (noon), 3DGS can consistently generate high-quality 3D models, with minimal impact of lighting variation on geometric accuracy and texture details.

TABLE II
3DGS Reconstruction Metrics under Varying Lighting Conditions (PSNR in dB, SSIM and LPIPS Dimensionless)

Dataset	PSNR	SSIM	LPIPS
Bank of China Tower-early morning	34.360	0.976	0.047
Bank of China Tower-noon	34.375	0.977	0.042
Bank of China Tower-dusk	35.186	0.975	0.049
Gardens by the Bay-early morning	35.815	0.981	0.033
Gardens by the Bay-noon	35.894	0.985	0.027
Gardens by the Bay-dusk	34.612	0.979	0.038
Tokyo Tower-early morning	34.803	0.979	0.033
Tokyo Tower-noon	34.596	0.981	0.030
Tokyo Tower-dusk	35.067	0.976	0.039

As indicated in Table 2, the quantitative metrics PSNR,

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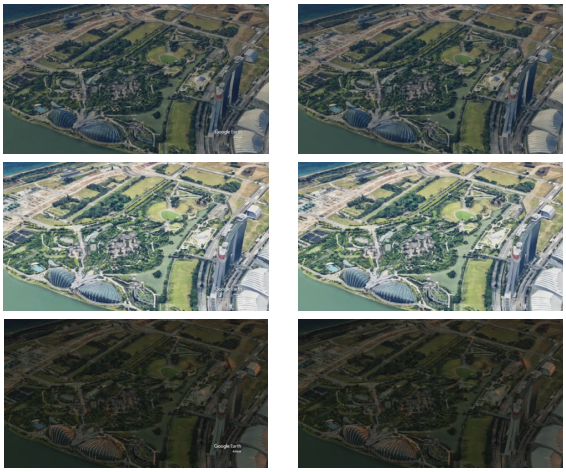

Ground Truth Reconstruction Results

Fig. 3. Urban scene reconstruction results using the 3DGS method under various lighting conditions. From top to bottom: early morning, noon, and dusk.

SSIM, and LPIPS not only exhibit excellent absolute values but also show very low variability between different time periods. For example, in the Gardens by the Bay dataset, the PSNR ranged from 34.612 dB to 35.894 dB, with a maximum variation rate of only 3.7%. The SSIM variation rate was below 0.6%, while the LPIPS variation was about 0.011, corresponding to a relative decrease of 28%. Similarly, in the Bank of China Tower and Tokyo Tower datasets, the maximum PSNR variation rate was below 3.5%, the SSIM variation rate was below 0.8%, and the LPIPS variation remained stable between 0.01–0.015. This slight fluctuation may stem from scene-specific characteristics: Gardens by the Bay features smoother vegetation textures and fewer reflective surfaces, whereas the other two sites contain more complex structures and fine edges, which may lead to higher perceptual dissimilarity. These results further prove that despite significant lighting changes between early morning, noon, and dusk, 3DGS is still able to adapt to complex lighting scenarios and maintain consistent modeling performance, demonstrating exceptional robustness.

V. Conclusion

This paper conducted large-scale urban 3D modeling based on the 3D Gaussian Splatting method and validated its potential applications in complex urban environments. Compared to the NeRF, 3DGS demonstrated superior performance in reconstruction accuracy and computational efficiency, with quantitative metrics including PSNR, SSIM, and LPIPS significantly outperforming NeRF, especially in handling complex scene boundaries and high-detail areas. Additionally, 3DGS showed good robustness under varying lighting conditions, with minimal impact on the model's geometric accuracy and texture details due to lighting changes. Overall, 3DGS shows great potential for digital twin modeling and urban planning support.

In future work, we plan to extend our validation to dense residential blocks and irregular street networks to further assess the generalization ability of 3DGS in complex and occluded urban scenarios. Although our input resolution (960×540) is relatively low, the model still achieves high-quality reconstruction; higher-resolution imagery will be explored in the future to enhance texture fidelity.

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